Adabala Cherisha Sesha Sai

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.20
LAKSHYA SCHOOL SAMALKOTA MANDAL EG DIST AP	Class XII (CBSE)	2024	94.80%
ADITYA (EM) HIGH SCHOOL, GANGARAJU NAGAR EAST GODAVARI DISTRICT	Class X (SSC)	2022	95.17%

POSITIONS OF RESPONSIBILITY

- **Operations Head**, EBSB, IIT Hyderabad(2026-Present)
- **Operations Coordinator**, EBSB, IIT Hyderabad (2025-26)
- **UG Mentor**, Sunshine, IIT Hyderabad (2025-26)

SKILLS

Technical Skills: Languages: Proficient in C++, Familiar with C, Python

Databases: PostgreSQL

Libraries / Tools: Pandas, Scikit-learn, Matplotlib, Git, GitHub.

Analytical & Soft Skills:

Applied analytical thinking and data interpretation in academic projects and DSA problem-solving

Coordinated institute-level events

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures and Applications, Algorithms, Database Management Systems, Theory of Computation, Discrete Maths for computer science

Mathematics & Machine Learning: Elementary Linear Algebra, Introduction to Artificial Intelligence, Calculus 1, Calculus 2, Introduction to Statistics, Probability and Random Variables, Differential Equations, Complex Variables, Transform Techniques

PROJECTS

In-Memory Cache Server (Mini-Redis) in C++ | Personal Project Github

- Implemented an in-memory key-value store combining a hash map with a doubly linked list to achieve O(1) time complexity for GET, SET, and LRU-based eviction
- Built a multi-threaded TCP server using socket programming with a custom RESP (REdis Serialization Protocol) parser, enabling concurrent clients to issue commands over a length-prefixed wire protocol
- Ensured thread-safe access to shared data structures using mutex locks to prevent race conditions during concurrent client requests
- Designed crash-safe persistence combining an append-only log with periodic atomic snapshotting (temp file + rename()), verified by hard-killing the process mid-session and confirming full data recovery on restart

Road Traffic Accident Analysis and Severity Prediction | personal project Github

- Tech Stack: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- Analyzed 12,316 road accident records from Addis Ababa using descriptive statistics and data visualization across

15 features including driver age, weather, and light conditions.
- Created 9 visualizations (bar charts, heatmaps, pie charts) to identify accident patterns across causes, junction types, and environmental conditions.
- Built and compared Logistic Regression and Random Forest classifiers to predict accident severity; Logistic Regression achieved **83.77% accuracy**, outperforming Random Forest (82.35%).
- Identified **Cause of Accident** and **Driving Experience** as the strongest predictors of severity using feature importance analysis.

URL Shortener REST API | personal project Github

- Built a REST API for URL shortening using FastAPI, with Base62 encoding for short-code generation and SQLite for persistent storage
- Designed endpoints for URL creation, redirection, and click-tracking analytics; used parameterized SQL queries to prevent injection vulnerabilities

EXTRACURRICULAR

- Actively participated in dance events since the age of 5.
- Secured **third prize** in the Kriya State-Level Inter-School Competitions for classical group dance.